

Tugas Praktikum Algoritma dan Struktur Data Pertemuan 10

Nama : Mochammad Fadil Aryan Siregar

NIM : J0403251161

Kelas : TPL A1

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# =====  
# Nama : Mochammad Fadil Aryan Siregar  
# Kelas : TPL A1  
# NIM : J0403251161  
# Pertemuan/praktikum 10  
# =====  
  
class Node:  
    def __init__(self, d):  
        self.left = None  
        self.right = None  
        self.data = d  
  
    def insert(self, d):  
        if self.data:  
            if d < self.data:  
                if self.left is None:  
                    self.left = Node(d)  
                else:  
                    self.left.insert(d)  
  
            if d > self.data:  
                if self.right is None:  
                    self.right = Node(d)  
                else:  
                    self.right.insert(d)  
        else:  
            self.data = d  
  
    def printTree(self):  
        if self.left:  
            self.left.printTree()  
        print(self.data)  
        if self.right:  
            self.right.printTree()  
  
def inorder(root):  
    return inorder(root.left) + [root.data] + inorder(root.right) if root else []  
  
def preorder(root):  
    return [root.data] + preorder(root.left) + preorder(root.right) if root else []  
  
def postorder(root):  
    return postorder(root.left) + postorder(root.right) + [root.data] if root else []  
  
# DATA NIM 56  
root = Node(56)  
data = [36, 76, 26, 46, 66, 86, 41]  
  
for d in data:  
    root.insert(d)  
  
print("=====  
print("NAMA : Mochammad Fadil Aryan Siregar")  
print("NIM : J0403251161")  
print("=====  
print("Inorder:", inorder(root))  
print("Preorder:", preorder(root))  
print("Post-order:", postorder(root))
```

1. Perbedaan ketiga cara traversal :

- Inorder : Kiri - Root - Kanan
- PreOrder : Root - Kiri - Kanan
- PostOrder : Kiri - Kanan - Root

2. Paling mudah untuk mencari root adalah PreOrder karena Root selalu muncul pertama

3. Hasil traversal akan berubah jika urutan berubah, cara traversalnya juga akan berubah